LEGO Design Enrichment Program: 8-Week Curriculum with Literacy

Think. Play. Grow. · Oct 3, 2026 · @Ariel Jasso

Daily Schedule (90 Minutes)

Every class follows the same five blocks, totaling 90 minutes of instruction plus recess. The weekly theme and Superpower stay the same as the original curriculum. Literacy happens every week, right after Morning Circle, and shows up again on the Builder Card during the competition.

Block	Time	What Happens
1. Morning Circle	15 min	Students settle in on the carpet or in a circle. Coach Ari introduces the weekly theme and Superpower with a short teaching presentation and a sample build or picture.
2. Read & Learn (Literacy)	10 min	Students read a short prepared passage aloud about the weekly theme and go over 3 vocabulary words. On research weeks, students use tablets to look up facts for their build.
3. Build Challenge	25 min	Students move to their seats and work on the weekly challenge, applying what they heard and read.
Recess	Not counted	Break.
4. Competition	25 min	Students come back from recess and go straight into the timed or judged competition. Each entry gets a Builder Card.
5. Cleanup & Prizes	15 min	8 minutes of sorting bricks back into bins, then 7 minutes for prizes, Superpower shout-outs, and one vocabulary word recap.
Total	**90 min**	

How the Literacy Component Works

Students read, speak, and write about the theme every week in two places: the 10-minute Read & Learn block and the Builder Card they fill out during competition.

Read & Learn (10 minutes)

- **Prepared passage.** Coach Ari brings a one-page reading about the week's theme. Suggested length: 100 to 150 words for Beginner, 150 to 200 for Intermediate, 200 to 300 for Advanced.
- **Reading aloud.** Use one format per week and rotate: echo reading (Coach reads a sentence, students repeat it), popcorn reading (students take turns reading one sentence each), or partner reading (pairs trade paragraphs).
- **Vocabulary.** Three bolded words per passage. Students say each word, hear a kid-friendly definition, and spot it in the reading.
- **One question.** End with one quick comprehension question, such as "What did the reading say makes a bridge strong?"

Tablet Research Weeks

Weeks marked "Tablet research" below replace the passage with 10 minutes of guided lookup. Students work in pairs and find two facts about their animal, vehicle, landmark, or time period. They write the facts on a sticky note and use at least one in their build. Keep the tablets on a bookmarked kid-safe search page or a short list of approved sites.

Builder Card (during Competition)

Every competition entry gets an index card with three lines:

1. Name of my build: ___
2. My build is ___ because ___. (Use one vocabulary word.)
3. My Superpower today was ___.

Beginners can draw and label, or dictate to a helper. Cards stay with the build for judging and the Week 8 gallery walk.

Beginner Level

Foundation skills: basic building techniques, structural stability, and creative expression. Beginner passages are short, and echo reading works best.

Week & Superpower	Morning Circle (15 min)	Read & Learn (10 min)	Build Challenge (25 min)	Competition (25 min)
Week 1: Introduction to LEGO Building Superpower: Organization	Show the main brick types (brick, plate, tile, slope) and how studs connect. Why sorted pieces	Passage: "What Is a LEGO Brick?" Vocab: stud, plate, stable.	Brick sorting and identification game, then color and shape pattern creation.	Tower building: tallest tower in 5 minutes, then a wobble test for most stable.

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	make building faster.			
Week 2: Building Strong Foundations Superpower: Persistence	Compare a straight stack to overlapped bricks, like a real brick wall. Push both to see which holds.	Passage: "How Builders Make Walls Strong." Vocab: foundation, overlap, weight.	Foundation strength tests and smooth walls with no visible studs.	Bridge building: span a gap between two books and hold the most weight.
Week 3: Symmetry & Patterns Superpower: Attention to Detail	Symmetry in butterflies, faces, and buildings. Demo with a small mirror.	Passage: "Symmetry All Around Us." Vocab: symmetry, pattern, mirror.	Mirror image building with a partner and pattern completion challenges.	Symmetrical creature creation, judged on how well the two sides match.
Week 4: Animals & Creatures Superpower: Imagination	Real animals vs. fantasy creatures. How eyes, tails, and wings tell you what kind of animal it is.	**Tablet research:** pairs find 2 facts about an animal or dinosaur. Vocab: habitat, feature, creature.	Zoo animal build that shows at least one fact they found.	Dragon or monster design challenge. Builder Card names one real animal feature they borrowed.
Week 5: Vehicles & Transportation Superpower: Problem-Solving	How wheels and axles work. Show a car that rolls and one that doesn't, and ask why.	Passage: "Wheels That Changed the World." Vocab: axle, wheel, motion.	Race car build, plus a boat or flying machine design.	Ramp race for distance, with a bonus boat float test.
Week 6: Buildings & Architecture Superpower: Planning	Parts of a building: doors, windows, roofs. Architects plan before they build; show a simple sketch.	Passage: "What Does an Architect Do?" Vocab: architect, blueprint, roof.	Sketch a dream house, then build it with interior details.	Store or shop building, then connect every build into one neighborhood.
Week 7: Storytelling with LEGO Superpower: Communication	Parts of a story: character, setting, problem. Model a quick scene.	Read a short prepared story (a fable works well) aloud together.	Recreate a favorite story or build your own story scene.	Mini-theater presentations. Each student reads a 2 to 3

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		Vocab: character, setting, plot.		sentence story caption aloud.
Week 8: Final Showcase & Celebration Superpower: Pride	Look back at the past 7 weeks and every Superpower earned.	Students read their favorite past Builder Card to a partner, then play a vocabulary review game.	Free build challenge with a theme: your personal masterpiece.	Gallery walk and presentations for peers and families. Each student reads their Builder Card. Awards ceremony during prizes.

Intermediate Level

Advancing skills: mechanical functions, advanced techniques, and collaborative building. Popcorn and partner reading work well at this level.

Week & Superpower	Morning Circle (15 min)	Read & Learn (10 min)	Build Challenge (25 min)	Competition (25 min)
Week 1: Advanced Building Techniques Superpower: Versatility	SNOT (Studs Not On Top) explained. Show brackets and a build with studs facing sideways.	Passage: "Building Sideways." Vocab: bracket, texture, orientation.	Sideways building challenges and texture experiments.	SNOT mosaic (a picture or initial), judged on clean, smooth surfaces.
Week 2: Simple Machines & Mechanisms Superpower: Curiosity	Gears, levers, and pulleys. Quick demo of each.	Passage: "Simple Machines Make Work Easier." Vocab: lever, gear, mechanical advantage.	Gear train construction and a pulley system challenge.	Catapult or trebuchet contest: hit the target from a set distance.
Week 3: Scale & Proportion Superpower: Precision	What scale means. Show a car that is too big for a minifigure and ask what went wrong.	**Tablet research:** look up the real size of a landmark or vehicle. Vocab: scale, proportion, ratio.	Minifigure-scale room or vehicle.	Landmark recreation at scale, judged on proportion and how recognizable it is.
Week 4: Collaborative Megastructures<b	Real cities are built by teams. Assign roles:	Partner reading: "Teams That Built Big." Vocab:	Team planning, then modular	Castle or fortress collaboration, judged on how

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r>Superpower: Teamwork	planner, builder, sorter, inspector.	collaborate, module, plan.	building sections for a team city.	well sections connect. Team Builder Card with one line from each member.
Week 5: Moving Creations Superpower: Innovation	Suspension and linkages. Push a stiff car and a sprung car over a bumpy surface.	Passage: "How Animals and Machines Walk." Vocab: suspension, linkage, kinetic.	Off-road vehicle with working suspension, plus a kinetic sculpture.	Walking creature race using linkages.
Week 6: Architecture & Interior Design Superpower: Vision	Interior design, modular buildings, and forced perspective (smaller things look farther away).	Passage: "Tricks Artists Use to Show Distance." Vocab: perspective, interior, modular.	Multi-story building with rooms, connected to a modular street system.	Forced perspective cityscape, judged from one marked viewing spot.
Week 7: Themed World Building Superpower: Creativity	What makes a world feel complete: matching colors, a clear theme, and a story.	**Tablet research:** 2 facts about a historical period, or ideas for a future city. Vocab: theme, era, environment.	Teams choose a fantasy realm, historical period, or future city and build several related structures.	World presentations. Each team reads a short "travel guide" paragraph from its Builder Card.
Week 8: Competition & Exhibition Superpower: Confidence	How to present like a professional. Coach models a 30-second presentation.	Students write and practice a short presentation script with a partner. Vocab: exhibit, judge, criteria.	Finish exhibition pieces and set up the exhibition.	Speed building competition, then gallery walk. Awards and reflection during prizes.

Advanced Level

Mastery: complex mechanisms, sculptural techniques, and engineering principles. Advanced students write more on their Builder Cards and lead parts of the reading.

Week & Superpower	Morning Circle (15 min)	Read & Learn (10 min)	Build Challenge (25 min)	Competition (25 min)
Week 1: Complex Mechanical Systems Superpower: Engineering Mindset	Multi-stage gears, gear ratios, and rack and pinion. Demo a small gear speeding up a big one.	Passage: "Inside a Clock." Vocab: gear ratio, rack and pinion, automaton.	Clock mechanism recreation and an automated lifting system.	Walking robot race to a finish line.
Week 2: Sculptural Building Techniques Superpower: Artistry	Making curves from square bricks. Show photos of large LEGO sculptures.	**Tablet research:** reference photos and 2 facts about the animal they will sculpt. Vocab: contour, organic, proportion.	Animal sculpture with realistic features, then a human figure.	Abstract sculpture challenge. Builder Card is written like a museum label.
Week 3: Structural Engineering Superpower: Analysis	Load, tension, and compression. Why triangles (trusses) make structures stronger.	Passage: "Why Bridges Stand Up." Vocab: load, tension, compression.	Cantilever design and an earthquake-resistant building tested on a shake table.	Bridge engineering weight test. Most weight held wins.
Week 4: Miniature & Microscale Superpower: Efficiency	Microscale examples. One brick can stand for a whole floor.	**Tablet research:** 2 facts about a famous landmark. Vocab: microscale, vignette, detail.	Group city in microscale plus a detailed vignette.	Famous landmark at mini scale. Classmates guess the landmark before reading the Builder Card.
Week 5: Kinetic Art & Motion Superpower: Synthesis	Artists who build moving art. Marble runs, cranks, and momentum.	Passage: "Can Anything Move Forever?" (why perpetual motion is impossible). Vocab: kinetic, crank, momentum.	Crank-operated animation and a "perpetual motion" sculpture that runs as long as possible.	Marble run with multiple paths. Longest run time wins.
Week 6: MOC Design Process Superpower: Design Thinking	MOC means "My Own Creation." Steps: sketch, gather pieces,	Read a real set of step-by-step building instructions aloud.	Original MOC: sketch, prototype, and write building instructions.	Instruction swap. A partner builds from your written instructions.

Week & Superpower	Morning Circle (15 min)	Read & Learn (10 min)	Build Challenge (25 min)	Competition (25 min)
	prototype, document.	Vocab: prototype, iterate, sequence.		Closest match wins.
Week 7: Master Builder Challenge Superpower: Resourcefulness	Real engineers work under limits on time, money, and materials.	Students read the week's design brief aloud like a real client request. Vocab: constraint, criteria, brief.	Specific theme with piece restrictions.	Mystery piece challenge, then a time-limited masterpiece.
Week 8: Portfolio & Exhibition Superpower: Mastery	What a portfolio is and what makes a good mentor.	Students write a short technical explanation of their best build and read it to a partner. Vocab: portfolio, mentor, technical.	Portfolio development using photos and past Builder Cards.	Exhibition with technical presentations, plus a helping-beginner-builders event if a younger group is available.

Weekly Prep Checklist

- ☐ Print the week's reading passage, one copy per student, with the 3 vocabulary words in bold
- ☐ Write one comprehension question for the end of Read & Learn
- ☐ Prepare the Morning Circle demo build or pictures
- ☐ Research weeks: charge tablets and load a bookmarked kid-safe search page
- ☐ Bring index cards for Builder Cards and sticky notes for research facts
- ☐ Set up judging criteria and prizes for the competition
- ☐ Write the vocabulary words on a whiteboard or poster for the prize-time recap

Open question: do the schools want reading passages matched to a specific grade level or reading standard? If so, passage length and vocabulary can be adjusted per site.